

BFHD

Bidirectional Feature Harmonization Decomposition for Heterogeneous Clinical Assessments

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Largest feasible measurement subsets in the original clinical spaces

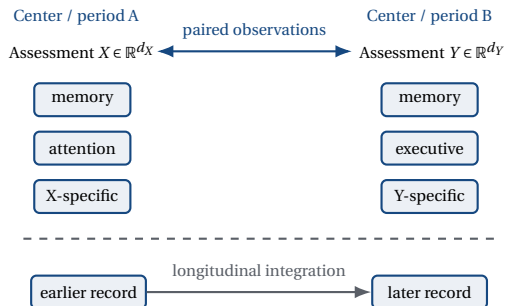
① Motivation

② Problem Formulation

③ Method

④ Results

Motivation



Clinical setting

Different centers, cohorts, and time periods often use related but non-identical assessment systems.

What integration must preserve

The harmonization must stay interpretable in the **original measurement spaces**, rather than only at the total-score level.

Clinical Question

Which original measurements can be reliably harmonized across protocols, and how can we maximize harmonization coverage without sacrificing reliability?

Existing approaches

Total-score linking

$$X_{total} \rightarrow Y_{total}$$

Only aligns total-score, does not identify recoverable measurements.

Predictive mappings

$$X \rightarrow Y \quad \text{or} \quad Y \rightarrow X$$

Usually target-specific and often predicts every target measurement without an explicit decomposition.

Representation learning

$$X \rightarrow Z \leftarrow Y$$

Aligns two views by mapping them into latent space, but does not return explicit subsets in the original measurement spaces.

Required output

A decomposition $(X_{\text{rec}}, X_{\text{nt}}, Y_{\text{rec}}, Y_{\text{nt}})$ with **maximal recoverable components**, **feature-wise reliability**, and **one bidirectionally coupled relationship**.

Dorans (2007); Kolen and Brennan (2013); Andrew et al. (2013); Zhu et al. (2017)

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PF step 1: maximize coverage under feature-wise feasibility

Clinical target

Find the largest subsets S_X and S_Y for which every retained measurement is ε -**recoverable**.

Feature-wise ε -recoverability

Let $I_X = \{1, \dots, d_X\}$ and $I_Y = \{1, \dots, d_Y\}$. For candidate mappings $f_\Phi : X \rightarrow Y$ and $g_\Phi : Y \rightarrow X$,

$$\mathbb{E}[d(X_i, g_\Phi(Y)_i)] < \varepsilon, \quad i \in S_X,$$

$$\mathbb{E}[d(Y_j, f_\Phi(X)_j)] < \varepsilon, \quad j \in S_Y.$$

$\varepsilon > 0$ specifies an application-level tolerance for acceptable agreement.

Largest feasible subsets

$$\max_{\substack{S_X \subseteq I_X, S_Y \subseteq I_Y, \\ \Phi \in \mathcal{M}}} w|S_X| + (1 - w)|S_Y|$$

$$\text{s.t.} \quad (S_X, S_Y, \Phi) \in \mathcal{F}_\varepsilon.$$

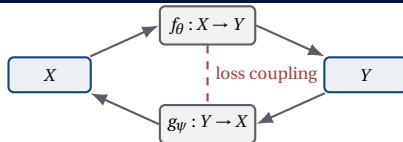
$\mathcal{F}_\varepsilon$ contains the triples for which every selected measurement satisfies the conditions on the left.

Output: explicit decomposition in the original measurement spaces

$$X_{\text{rec}} = X_{S_X^*}, \quad X_{\text{nt}} = X_{I_X \setminus S_X^*}, \quad Y_{\text{rec}} = Y_{S_Y^*}, \quad Y_{\text{nt}} = Y_{I_Y \setminus S_Y^*}.$$

PF step 2: recoverability is relative to a coupled bidirectional model class

Two separately parameterized mappings



The two directions may still drift because θ and ψ can change separately.

Bidirectionally coupled model class $\mathcal{M}$



Updates to $X \rightarrow Y$ must simultaneously constrain, and be constrained by, $Y \rightarrow X$.

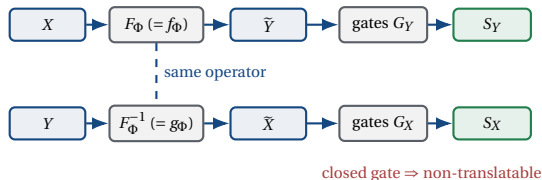
Why this restriction matters

Optimizing one direction induces corresponding updates in the other, rather than allowing two unrelated functions linked only by loss terms.

The population expectations and subset search are not directly tractable; BFHD provides the practical relaxation.

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BFHD turns the formulation into a trainable decomposition



Direct correspondence to the PF

single operator $\leftrightarrow \mathcal{M}$; output gates $\leftrightarrow (S_X, S_Y)$;
 hinge feasibility $\leftrightarrow \mathcal{F}_\varepsilon$; gate reward $\leftrightarrow$ maximal coverage.

Main implementation: a NICE-style invertible backbone.
 Invertibility enforces a single operator; it is not an assumption that the clinical systems are invertible.

Feasibility–coverage objective

$$\begin{aligned} \mathcal{L}_{\text{total}} = & \sum_i G_{X,i} \text{ReLU}[\text{err}(X_i, \tilde{X}_i) - \varepsilon_{\text{train}}] \\ & + \sum_j G_{Y,j} \text{ReLU}[\text{err}(Y_j, \tilde{Y}_j) - \varepsilon_{\text{train}}] \\ & - \lambda_{L0} \left(\sum_i G_{X,i} + \sum_j G_{Y,j} \right). \end{aligned}$$

- $\text{err}(\cdot, \cdot)$ is feature-wise NMSE.
- The hinge terms penalize only violations of $\varepsilon_{\text{train}}$.
- The gate reward keeps as many gates open as possible.

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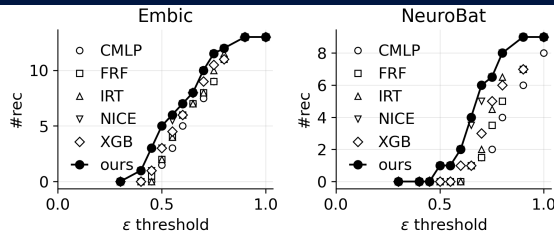
The experiments test the proposed decomposition

Synthetic: known recoverability labels

Model	Acc_X	Acc_Y	J_X	J_Y
CMLP	0.855	0.810	0.975	0.789
NICE	0.735	0.725	0.837	0.814
BFHD	0.880	0.830	0.933	0.924

BFHD gives the best overall index-level structure recovery, with strong subset stability.

Real data: coverage under the same feasibility tolerance



Across the intermediate ϵ regime, BFHD (“ours”) identifies broader feasible subsets under the same criterion.

The selected decomposition remains feasible under protocol drift

Temporal deployment protocol



no re-selection on the later cohort

EMBIC(A) $\leftrightarrow$ NeuroBAT(B), $\varepsilon_{\text{cutoff}} = 0.6$

Method	#Ret _A	Ret% _A	#Ret _B	Ret% _B
XGB	6	85.7	0.5	62.5
CMLP	6	71.0	1	90.0
NICE	6	77.4	2	49.2
BFHD	7	100.0	3	100.0

Conclusion

- ① Feature harmonization should return a maximal feasible decomposition in the original measurement spaces.
- ② Recoverability should be defined under one coupled bidirectional model class.
- ③ BFHD yields broader, more stable, and more deployable harmonization decisions.

Thank you

Questions?